Minimal surfaces

Exercise sheet 2

Exercise 1 (Basics of spectral theory)**.** Let (Σ, g) be a connected, smooth, compact Riemannian manifold without boundary,[1] and let $q \in C^\infty(\Sigma)$ be a smooth function on Σ.

We are interested in the eigenvalue problem associated to the Schrödinger operator $J = \Delta + q$. Namely, we are looking for nonzero solutions of

$$-Ju = \lambda u.$$

In this case, λ is called *eigenvalue* and u is called *eigenfunction* associated to eigenvalue λ.

For $u, v \in H^1(\Sigma) = W^{1,2}(\Sigma)$, define

$$Q(u,v) = \int_\Sigma \big(\langle \nabla u, \nabla v\rangle - quv\big)\,\mathrm{dvol}_g.$$

Moreover, for $u \in H^1(\Sigma) \setminus \{0\}$, we define the *Rayleigh quotient*

$$R(u) = \frac{Q(u,u)}{\int_\Sigma u^2\,\mathrm{dvol}_g}.$$

Background results. The following results may be used without proof.

(i) **Lax–Milgram theorem.** A continuous, coercive bilinear form on a Hilbert space gives a unique weak solution to every continuous linear functional.

(ii) **Rellich–Kondrachov theorem.** The embedding $H^1(\Sigma) \hookrightarrow L^2(\Sigma)$ is compact.

(iii) **Spectral theorem.** A compact self-adjoint operator on a Hilbert space has an orthonormal basis of eigenvectors (including its kernel). Its nonzero eigenvalues have finite multiplicity and can accumulate only at 0.

(iv) **Elliptic regularity.** Every H^1 weak solution of $-Ju = \lambda u$ is smooth.

(v) **Strong minimum principle for $\Delta - C$.** Let Ω be a connected open subset of a Riemannian manifold, let $C > 0$, and suppose $u \in C^2(\Omega)$ satisfies

$$(\Delta - C)u \le 0.$$

If u attains a nonpositive minimum at an interior point of Ω, then $u \equiv 0$ in Ω.

(vi) **Unique continuation.** If a smooth solution of $-Ju = \lambda u$ vanishes on a nonempty open set, it vanishes identically on connected Σ.

Part I. Discreteness of the spectrum. Choose $c > \|q\|_{L^\infty} + 1$ and set

$$Q_c(u,v) = Q(u,v) + c\int_\Sigma uv\,\mathrm{dvol}_g.$$

(a) Prove that Q_c is continuous and coercive on $H^1(\Sigma)$. Deduce that, for each $f \in L^2(\Sigma)$, there is a unique $u \in H^1(\Sigma)$ such that

$$Q_c(u,v) = \int_\Sigma fv\,\mathrm{dvol}_g \qquad \text{for every } v \in H^1(\Sigma).$$

Denote this solution by Tf. Show that $T\colon L^2(\Sigma) \to H^1(\Sigma)$ is bounded.

[1] Many of the results here work on a manifold with boundary, where we study the Schrödinger operator J with suitable boundary conditions (e.g., Dirichlet or Neumann). We study here the case without boundary for simplicity.

(b) Regard T as an operator $L^2(\Sigma) \to L^2(\Sigma)$. Prove that T is compact, self-adjoint, injective, and strictly positive in the sense that $\langle Tf, f\rangle_{L^2} > 0$ for $f \neq 0$.

(c) Apply the spectral theorem to obtain an $L^2(\Sigma)$-orthonormal basis $(\varphi_j)_{j\geq 1}$and numbers

$$\mu_1 \geq \mu_2 \geq \ldots > 0, \qquad \mu_j \longrightarrow 0, \qquad T\varphi_j = \mu_j \varphi_j.$$

Show that φ_j is smooth and satisfies

$$J\varphi_j + \lambda_j \varphi_j = 0, \qquad \lambda_j = \mu_j^{-1} - c.$$

Conclude that the eigenvalue problem $-Ju = \lambda u$ has a discrete sequence of real eigenvalues

$$\lambda_1 \leq \lambda_2 \leq \ldots \longrightarrow +\infty,$$

listed with multiplicity, each of finite multiplicity.

Part II. Variational characterization. Prove the following variational characterizations of the eigenvalues.

(a) Show that $\left(\varphi_j/\sqrt{\lambda_j + c}\right)_{j\geq 1}$ is an orthonormal basis of $H^1(\Sigma)$ for the inner product Q_c. Deduce that for every $u \in H^1(\Sigma)$, defining $u_j = \int_\Sigma u\varphi_j \,\mathrm{dvol}_g$, we have

$$\int_\Sigma u^2 \,\mathrm{dvol}_g = \sum_{j=1}^{\infty} u_j^2, \qquad Q(u,u) = \sum_{j=1}^{\infty} \lambda_j u_j^2.$$

(b) Prove the Rayleigh characterization of the first eigenvalue:

$$\lambda_1 = \min_{u \in H^1(\Sigma)\setminus\{0\}} R(u).$$

More generally, show that

$$\lambda_k = \min_{\substack{u \in H^1(\Sigma)\setminus\{0\} \\ u \perp \varphi_1, \ldots, \varphi_{k-1}}} R(u),$$

where orthogonality is in $L^2(\Sigma)$.

(c) Prove the *min–max characterization*:

$$\lambda_k = \min_{\substack{V \subset H^1(\Sigma) \\ \dim V = k}} \max_{u \in V\setminus\{0\}} R(u).$$

Part III. Positivity and simplicity of the first eigenvalue. Prove that the first eigenfunction can be chosen strictly positive and that the first eigenvalue is simple.[2]

Part IV. Courant's nodal domain theorem. Prove that an eigenfunction φ_k relative to eigenvalue λ_k has at most k nodal domains.[3]

Hint: Let $\Omega_1, \ldots, \Omega_n$ be the nodal domains of φ_k. Consider

$$u_i = \begin{cases} \varphi_k(x), & x \in \Omega_i, \\ 0, & x \notin \Omega_i. \end{cases}$$

Then $u_i \in H^1(\Sigma)$ and $\nabla u_i = \mathrm{id}_{\Omega_i} \nabla \varphi_k$ almost everywhere.

[2] Meaning that the first eigenspace has dimension one.

[3] A nodal domain of φ_k is a connected component of $\{x \in \Sigma : \varphi_k(x) \neq 0\}$.